

Signal Processing Decision Pipeline

Chronological Architecture & Method Selector

Stage 1: Signal Acquisition & Digitization

Core Purpose: Capture physical physiological voltages and convert them into uncorrupted digital time series data.

Relevant Tools & Methods:

Ag/AgCl Electrodes

Anti-Alias Filtering (Analog Low-Pass)

Nyquist-Shannon Sampling

Analog-to-Digital Conversion (ADC)

Quantization

Non-Uniform Resampling & Interpolation

Key Tool Choice Question: Is our sampling frequency at least double the highest frequency component of our signal to prevent aliasing?

Stage 2: Domain Transformation & Representation

Core Purpose: Re-express raw waveforms in time, frequency, or scale domains to reveal hidden physiological rhythms and patterns.

Relevant Tools & Methods:

Fast Fourier Transform (FFT)

Short-Time Fourier Transform (STFT / Spectrogram)

Tapered Window Functions (Hann, Hamming, Blackman, Kaiser)

Continuous & Discrete Wavelet Transforms (CWT, DWT / Scalogram)

Hilbert Transform (Analytic Signal & Instantaneous Envelope)

Discrete Cosine Transform (DCT)

Autoregressive (AR) Parametric Modeling

Welch's Method & Multitaper Spectral Estimation

Key Tool Choice Question: Is our signal stationary across the epoch, or do we need time-frequency wavelets to capture rapid transient events?

Stage 3: Noise & Artifact Removal

Core Purpose: Eliminate specific noise signatures and interference without distorting underlying physiological components.

Relevant Tools & Methods:

Median Filter

Notch Filter (50/60 Hz)

High-Pass Filter & Linear Detrending

FIR & IIR Digital Filters (Butterworth, Chebyshev, Elliptic)

Zero-Phase Filtering (filtfilt)

Savitzky-Golay Filter

Ensemble Averaging

Least Mean Squares (LMS) Adaptive Cancellation

Wavelet Coefficient Thresholding (Hard / Soft Shrinkage)

ICA & PCA

Key Tool Choice Question: What is the precise signature of the contamination (impulsive spike, powerline hum, slow drift, or in-band noise), and which tool targets it directly?

Stage 4: Event & Component Detection

Core Purpose: Pinpoint and isolate specific physiological boundaries, wave peaks, or graphoelements in continuous time.

Relevant Tools & Methods:

Pan-Tompkins Pipeline (Band-Pass, Derivative, Squaring, Integration, Adaptive Threshold)

Five-Point Derivative & Slope High-Pass

Moving-Window Integration

Template Matching & Cross-Correlation

Peak & Threshold Detection

Key Tool Choice Question: Are we locating sharp, discrete repeating events (like heartbeats) or segmenting continuous physiological states?

Stage 5: Feature Extraction & Selection

Core Purpose: Summarize variable signal windows into compact, fixed-length numerical vectors that separate target classes.

Relevant Tools & Methods:

Time-Domain Features (RMS, MAV, Variance, Zero-Crossing Rate)

Hjorth Parameters (Activity, Mobility, Complexity)

Sample Entropy

Frequency-Domain Features (Band Power, Relative Power, Spectral Edge, Spectral Entropy)

Morphological & Fiducial Features (P-QRS-T Intervals & Amplitudes)

Common Spatial Patterns (CSP)

Feature Standardisation (In-Fold Z-Scoring)

Mutual Information Ranking

ANOVA F-Test

Lasso L1 Regularization & Random Forest Importance

Key Tool Choice Question: Which concise set of features captures the key physiological differences without introducing redundancy or the curse of dimensionality?

Stage 6: Classification & Pattern Recognition

Core Purpose: Train mathematical decision models to map extracted feature vectors or learned representations to target clinical labels.

Relevant Tools & Methods:

k-Nearest Neighbors (k-NN)

Support Vector Machines (SVM)

Decision Trees & Random Forests

1D & 2D Convolutional Neural Networks (CNNs)

Sequence Models (LSTM, Transformers)

Self-Supervised Time-Series Foundation Models

Class Imbalance Oversampling (SMOTE / ADASYN)

Key Tool Choice Question: Do we have sufficient labeled data for deep learning, or does a classical model like a Random Forest provide better accuracy and interpretability?

Stage 7: System Validation & Deployment

Core Purpose: Verify system performance without data leakage and package the frozen pipeline for real-world or edge deployment.

Relevant Tools & Methods:

Leave-One-Subject-Out (LOSO) Cross-Validation

GroupKFold Splitting

In-Fold Transformation Fitting

Cohen's Kappa & Macro-F1 (with Spread)

Confusion Matrices & Bland-Altman Plots

TinyML & On-Device Edge Optimization

Frozen Inference Pipeline (predictions.csv)

Key Tool Choice Question: Has every parameter-learning step been strictly isolated inside cross-validation folds to ensure zero test-set leakage?